E-Commerce Intelligent Recommendation System

Hybrid AI + Machine Learning + Multi-Agent Architecture

Project objective

This project builds an intelligent system that recommends products to users in two real-world scenarios:

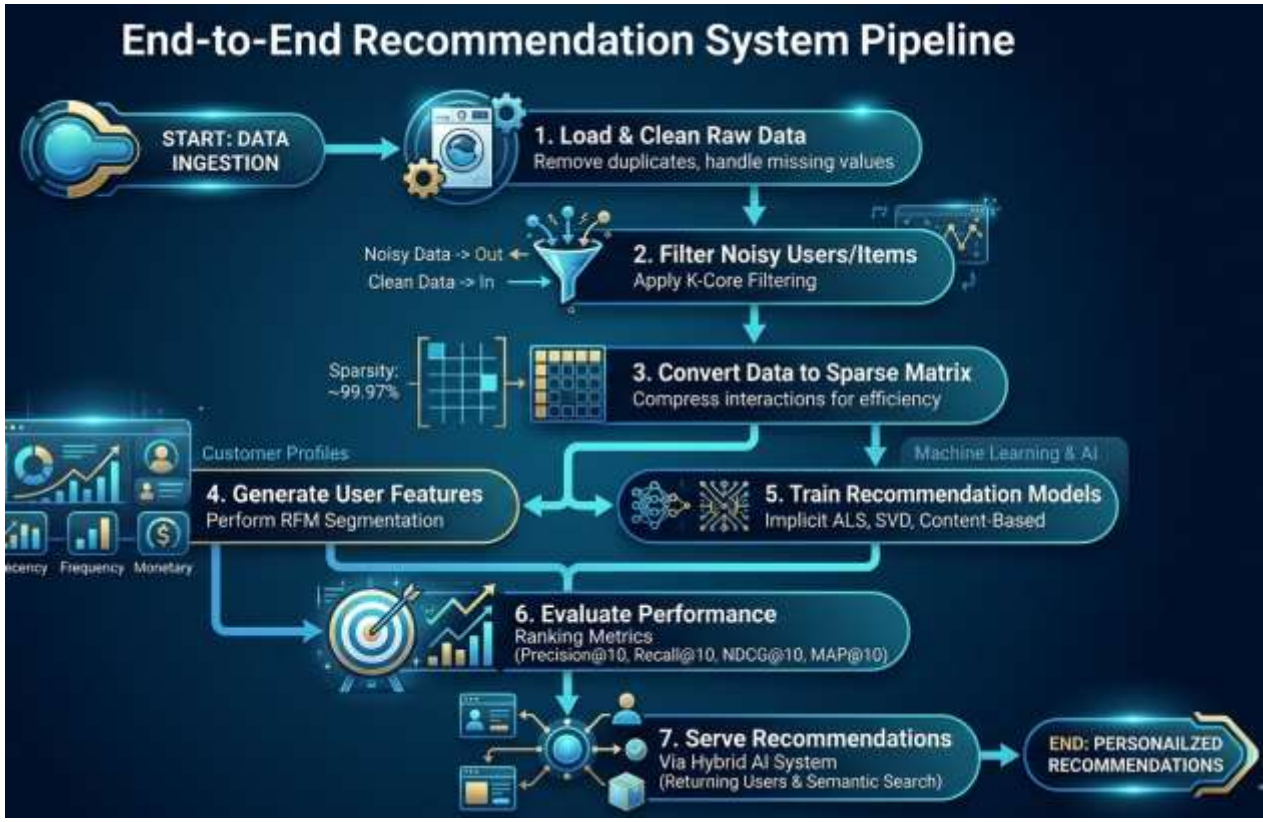
- A returning user who expects personalized suggestions
- A new user who searches using natural language

Instead of relying on a single model, the system combines multiple AI techniques and intelligently decides **how to recommend**, not just **what to recommend**.

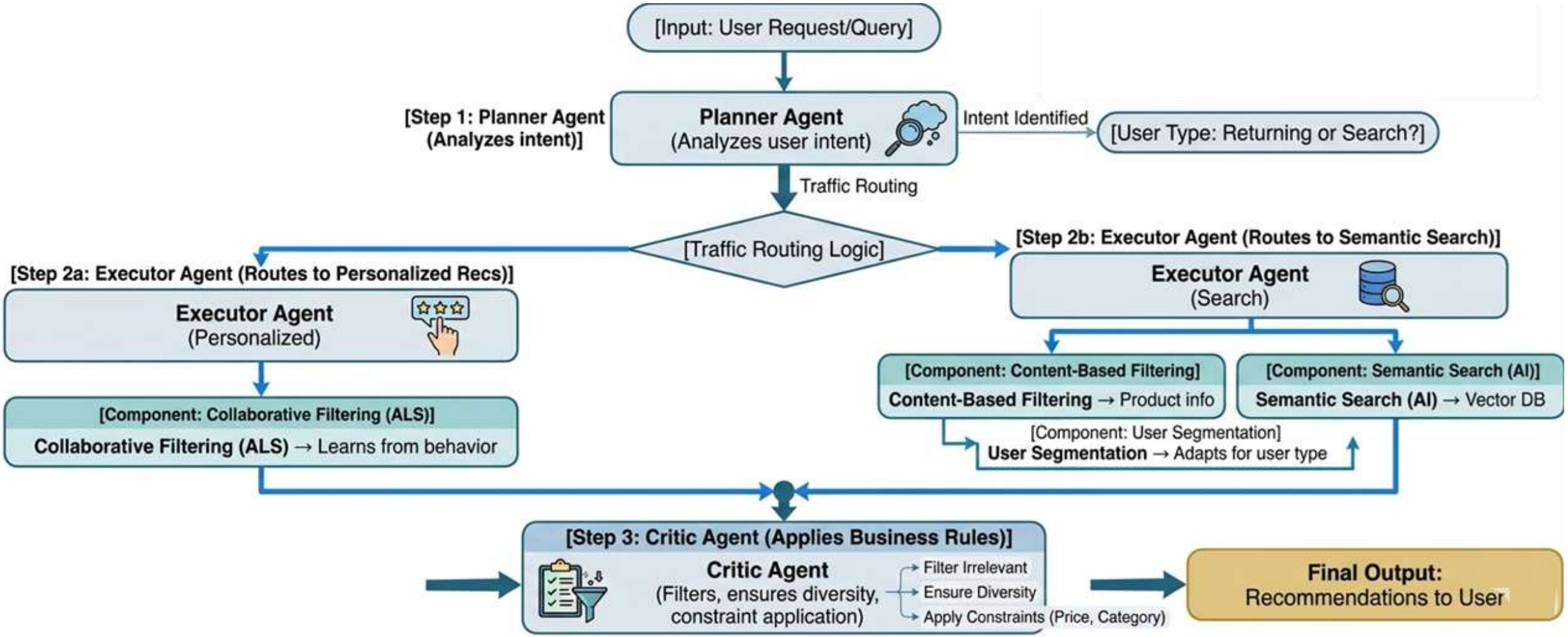
Project Structure

```
src/
 ├── data/          → Data loading & preprocessing
 ├── features/      → Feature engineering (RFM, embeddings)
 ├── models/        → ML models (ALS, SVD, content-based, hybrid)
 ├── evaluation/    → Metrics (Precision, Recall, NDCG, MAP)
 └── utils/         → Supporting utilities

data/
 ├── raw/           → Original dataset
 ├── processed/     → Cleaned data
 ├── features/      → Model-ready features

artifacts/          → Saved models & mappings
app.py              → Streamlit application
main.py             → End-to-end pipeline
```

Project pipeline



Multi-Agent Pipeline



Evaluation Strategy

Instead of random splitting, the system uses:

Temporal Split (Realistic Evaluation)

•Train on past data

•Predict future interactions

Metrics Used:

•Precision@10 → how accurate recommendations are

•Recall@10 → how many relevant items are found

•NDCG@10 → ranking quality

•MAP@10 → overall ranking performance

Results

•ALS outperformed all baseline models

•Hybrid system improved coverage and flexibility

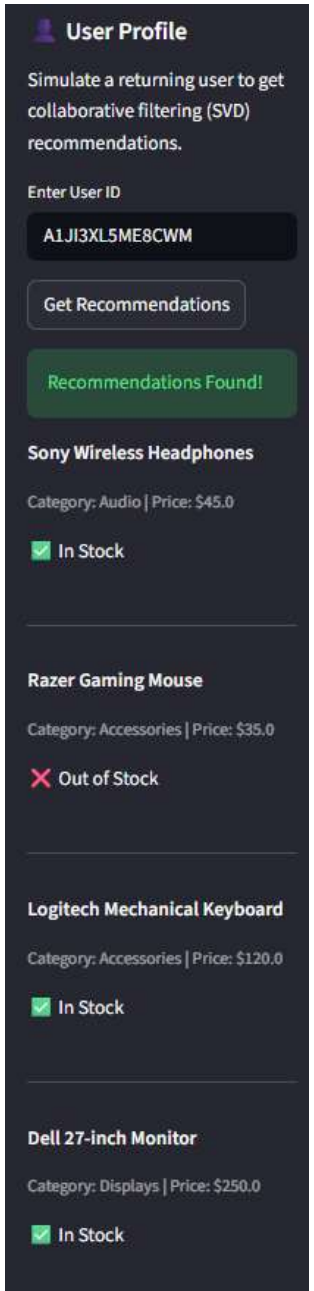
•Content-based model helped solve cold-start problem

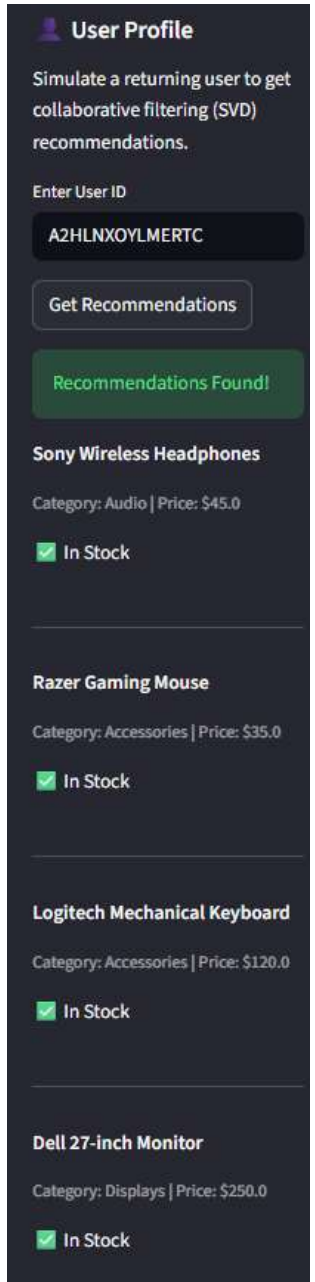
Key takeaway:

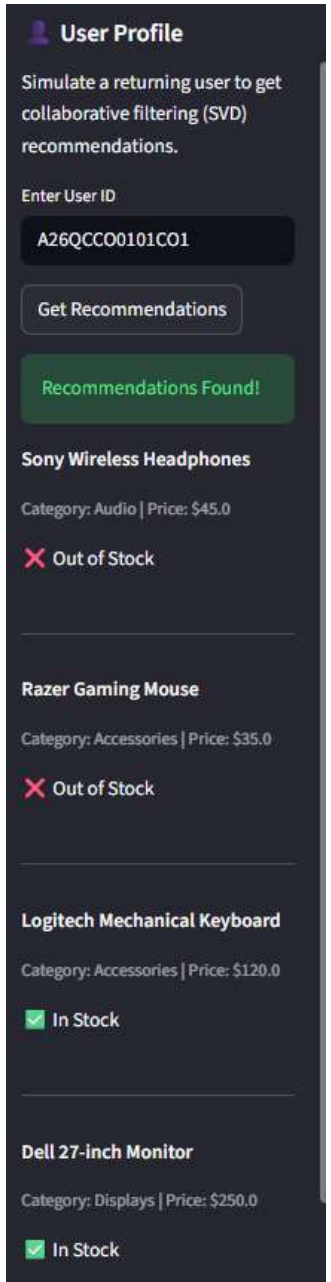
→ No single model is sufficient

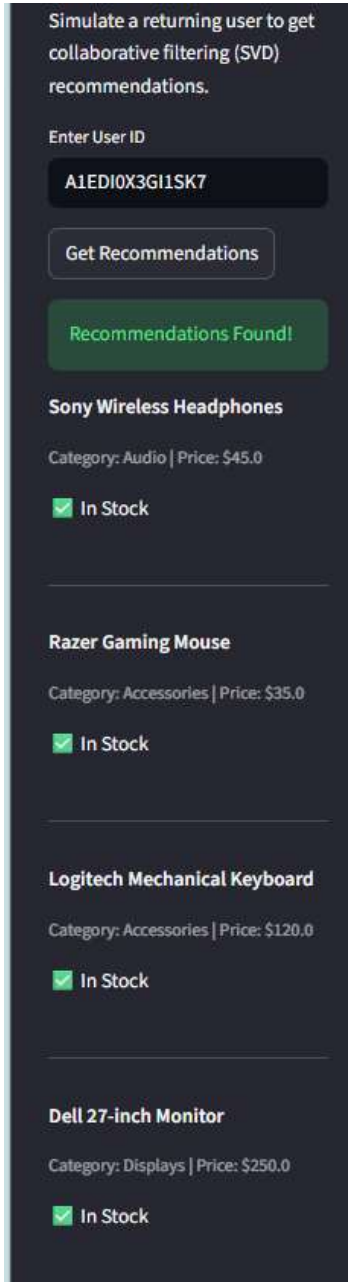
→ Hybrid systems perform best in real-world scenarios

Dashboard Overview









Dashboard Overview

🛒 AI-Powered E-Commerce Recommendation Engine

Built by **Geetha** | Powered by SVD Collaborative Filtering & ChromaDB RAG

🔍 Semantic Product Search (RAG)

Search for products using natural language. The AI will understand intent and budget!

What are you looking for?

gaming mouse with RGB lighting

Search Database

Top 5 AI Matches from Catalog:

Result #1

Category: Unknown | Price: $7.99

Product: Satechi Spectrum Mouse Wired Optical 7-Color LED Changing Computer Mouse with 1000 DPI (Sil.....

View Product

Result #2

Category: Unknown | Price: $0.0

Product: E-Blue Mazer Type-R 2400DPI USB Wired Optical Gaming Mouse. Category: Unknown. Price: 0.0. .....

View Product

Result #3

Category: Unknown | Price: $6.29

Product: Logitech 931179-0403 MX510 Optical Mouse RED. Category: Unknown. Price: 6.29. Description: .....

View Product

Result #4

Category: Unknown | Price: $0.0

Product: HDE Ultra-Thin Wireless Mouse 2.4GHZ Ergonomic Candy Colored Optical Mouse with USB Nano Re.....

View Product

Result #5

Category: Unknown | Price: $5.6

Product: Importer520 Black 3-Button 3D USB 800 Dpi Optical Scroll Mice Mouse w/Red LEDs For Notebook.....

View Product

CHALLENGES	SOLUTIONS
1. Extreme Data Sparsity Most users interact with very few items. Real-world data is extremely incomplete. (~99.97% of interactions missing)	**1. Matrix Optimization and Filtering** Applied K-Core filtering to reduce noise. Used a Sparse Matrix representation for computational efficiency.
2. Cold Start Problem New users or new products lack historical interaction data. (This prevents meaningful personalization).	**2. Content-Based + Semantic Models** Built Content-Based and Semantic Search (AI) models that understand product title, category, and description text to handle new items.
3. MESSY Real-World Data Inconsistent product metadata, missing descriptions, and missing prices.	**3. Hybrid Architecture and Preprocessing** Designed a hybrid architecture instead of relying on a single model. Applied explicit data loading and cleaning (data preprocessing) to raw data.
4. Slow Recommendation Generation Generating recommendations at large-scale computation (150K+ users) is computationally expensive.	**4. Sparse Matrix + Scalable Solvers** Used Implicit ALS (Alternating Least Squares), which is designed for sparse implicit data and large-scale, efficient computation.
5. Generic Recommendations Most platforms show generic products (e.g., just popular items) to all users, resulting in poor user engagement.	**5.Multi-Agent Pipeline & User Segmentation** **5.** Introduced an AI-driven Multi-Agent decision layer. The system generates user features (RFM segmentation) to adapt recommendations based on the user type (returning user vs. search query).

Impact

Business Impact

•Improved personalization → higher conversion potential

•Handles both browsing and search use cases

•Scalable to real-world datasets

Technical Impact

•Demonstrates production-grade system design

•Combines ML + NLP + system architecture

•Mimics real-world recommendation engines

Future Improvements

•Use **ranking-based models (BPR)**

•Add **real-time recommendation API**

•Improve hybrid re-ranking logic

•Deploy using cloud infrastructure